Vidhi Jain

✉ vidhij@alumni.cmu.edu
vidhijain.github.io
LinkedIn: vidhijain96/
ORCID: 0009-0006-8170-4115

Education

2022–2025 **Ph.D. in Robotics from Carnegie Mellon University (Robotics Institute)**.
Robotics Institute, Advisor: Yonatan Bisk

2019–2021 **Masters of Science in Robotics, Carnegie Mellon University**.
Robotics Institute, *GPA 4.0* Advisor: Katia Sycara

2014–2018 **Bachelor of Engineering (Honors) in Computer Science, BITS Pilani**.
CGPA 9.06/10.00, Distinction, Advisor (off-campus thesis): Aaron Courville

Experience

Jul'25–Present **OpenAI**, *San Francisco, CA, USA*.
Researcher in AI transformation team with Boris Power
- Engineered novel evaluation, synthetic data pipelines, ML training pipelines and showed measurable LLM performance gains in the chip design domain.
- Led the scaling of research prototypes for large-scale RL training, demonstrating consistently improved performance gains with increased inference compute and model capacity.

Jun'23–Feb'24 **Google DeepMind**, *Mountain View, CA, USA*.
Student Researcher with Debidatta Dwibedi
- Trained video-conditioned robot policies with cross-attention transformers. Vid2Robot (RSS'24)
- Extended models for visual grounding with conditional caption generation for QA. FlexCap (NeurIPS'24)
- Evaluated LLM prompts for low-level motion control for articulated object interaction. PromptBook (ICRA'24)
- Developed pipelines for zero-shot and fine-tuned task evaluations for single neural network policy trained on aggregated robotics data. Open X-Embodiment (ICRA'24) Best Paper.

Sep'21–Aug'22 **Meta (Facebook)**, *Menlo Park, CA, USA*.
AI resident with Akshara Rai and Yixin Lin
- Developed synthetic data for robot demonstrations of loading the simulated dishwasher in AI Habitat. Code
- Trained Encoder-decoder Transformer-based high-level policy for learning preferences in loading dishwasher from a single demonstration as prompt. Paper | Website

Aug'18–Aug'19 **Microsoft Research**, *Bengaluru, India*.
Research Fellow with Amit Deshpande and Navin Goyal
- **Divergence minimization in GANs** Investigated the properties of Jensen Shannon and Wasserstein divergences between the given and the generated distributions during training neural network generators, to discuss the learning dynamics like convergence and generalization.

Jan'18–Jun'18 **Montreal Institute for Learning Algorithms (MILA)**, *Montreal, QC, Canada*.
Research Intern with Aaron Courville
- **Out-of-Distribution Detection in Generative Models** Experimented on autoregressive generative models and Variational Autoencoders, to detect any 'out-of-distribution (OOD)' sample at deployment. Thesis

May'17–Jul'17 **Simon Fraser University (SFU)**, *Burnaby, BC, Canada*.
MITACS Globalink Research Intern with Oliver Schulte
- **Learning Bayesian Networks for Relational Databases** Remodelled existing codebase for project *FactorBase* to integrate cross-table sufficient statistics or population variables in the contingency tables for learning ground-level sufficient statistics. Code | Invited Paper

May'16–Jul'16 **Forschungszentrum Informatik (FZI)**, *Karlsruhe, Germany*.
Research Intern with York Sure-Vetter and Matthias Frank

- **API annotation platform using Semantic Web** Designed and developed a customized Semantic MediaWiki for the decision support system of Project *BigGIS*; Extended Abstract | Poster

Publications

Aug 2024 **ANAVI: Audio Noise Awareness using Visuals of Indoors for NAVIgation**.
Vidhi Jain, Rishi Veerapanini, Yonatan Bisk. *In Proceedings of* Conference of Robot Learning (CoRL) 2024, Munich, Germany. Paper | Website

Mar 2024 **FlexCap: Generating Rich, Localized, and Flexible Captions in Images**.
Debidatta Dwibedi, **Vidhi Jain**, Jonathan Tompson, Andrew Zisserman, Yusuf Aytar. *Accepted at* Conference on Neural Information Processing Systems (NeurIPS) 2024, Vancouver, Canada. Paper | Website

Jan 2024 **Vid2Robot: End-to-end Video-conditioned Policy Learning with Cross-Attention Transformers**.
Vidhi Jain, Maria Attarian, Nikhil J Joshi, Ayzaan Wahid, Danny Driess, Quan Vuong, Pannag R Sanketi, Pierre Sermanet, Stefan Welker, Christine Chan, Igor Gilitschenski, Yonatan Bisk, Debidatta Dwibedi, *In Proceedings of* Robotics: Science and Systems (RSS) 2024, Delft, Netherlands. Paper | Website | Video

Nov 2023 **How to Prompt Your Robot: A PromptBook for Manipulation Skills with Code as Policies**.
Montserrat Gonzalez Arenas, Ted Xiao, Sumeet Singh, **Vidhi Jain**, Allen Z. Ren, Quan Vuong, Jake Varley, Alexander Herzog, Isabel Leal, Sean Kirmani, Dorsa Sadigh, Vikas Sindhwani, Kanishka Rao, Jacky Liang, Andy Zeng. *In Proceedings at* International Conference on Robotics and Automation (ICRA) 2024, Yokohama, Japan. Paper

Nov 2023 **Open X-Embodiment: Robotic Learning Datasets and RT-X Models**.
Open X-embodiment collaboration. *In Proceedings at* International Conference on Robotics and Automation (ICRA) 2024, Yokohama, Japan. Best Paper Paper | Website

Jun 2023 **SLAP: Spatial-Language Attention Policies**.
Priyam Parasher, **Vidhi Jain**, Xiaohan Zhang, Jay Vakil, Sam Powers, Yonatan Bisk, Chris Paxton. *In Proceedings of* Conference on Robot Learning (CoRL) 2023, Atlanta, USA., Paper | Website

Jun 2023 **HomeRobot: Open-Vocabulary Mobile Manipulation**.
Sriram Yenamandra, Arun Ramachandran, Karmesh Yadav, Austin S Wang, Mukul Khanna, Theophile Gervet, Tsung-Yen Yang, **Vidhi Jain**, Alexander Clegg, John M Turner, Zsolt Kira, Manolis Savva, Angel X Chang, Devendra Singh Chaplot, Dhruv Batra, Roozbeh Mottaghi, Yonatan Bisk, Chris Paxton. *In Proceedings of* Conference on Robot Learning (CoRL) 2023, Atlanta, USA. Paper | Website | Competition @ NeurIPS 2023

Dec 2022 **Transformers are Adaptable Task Planners**.
Vidhi Jain, Yixin Lin, Eric Undersander, Yonatan Bisk, Akshara Rai. *In Proceedings of* Conference on Robot Learning (CoRL) 2022, Auckland, New Zealand. Paper | Website | Video | Code

Workshop Presentations

Dec 2022 **MAEA: Multimodal Attribution for Embodied AI**.
Vidhi Jain, Jayant Sravan Tamarapalli, Sahiti Yerramilli, Yonatan Bisk. *Presented at* NeurIPS 2022, 5th Robot Learning Workshop: Trustworthy Robotics (RLW 2022), New Orleans, USA. Short Paper | Website | Video

Oct 2020 **Predicting Strategies in Simulated Search and Rescue**.
Vidhi Jain, Rohit Jena, Huao Li, Tejus Gupta, Dana Hughes, Michael Lewis, Katia Sycara. *Presented at* NeurIPS AI+HADR 2020, Virtual. Video | Preprint

Oct 2020 **Learning Embeddings that Capture Spatial Semantics for Indoor Navigation**.
Vidhi Jain, Shishir Patil, Prakhar Agarwal, Katia Sycara. *Presented at* NeurIPS 2020 Object Representations for Learning and Reasoning (ORLR 2020), Virtual. Video | Preprint | Code

Sep 2020 **Learning to Navigate in Unseen Cluttered Structured Environments**.
Vidhi Jain, Ganesh Iyer, Katia Sycara. NeurIPS 2020 Women In Machine Learning (WiML 2020), Virtual. Poster

Jul 2020 **Coping with Sample Inefficiency in Deep Reinforcement Learning (DRL) in embodied AI**.
Vidhi Jain, Simin Liu, Ganesh Iyer. ICML'20 Women In Machine Learning (WiML) Un-Workshop, Virtual. Discussion | Presentation

Aug 2017 **Model Selection Scores for Multi-Relational Bayesian Networks**.
Sajjad Gholami, Oliver Schulte,**Vidhi Jain**, Qiang Zhao. IJCAI 2017 Declarative Learning Based Programming (DeLBP). Invited Paper

Preprints

Dec 2023 **Towards General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis**.
Yafei Hu*, Quanting Xie*, **Vidhi Jain***, Jonathan Francis, Jay Patrikar, Nikhil Keetha, Seungchan Kim, Yaqi Xie, Tianyi Zhang, Shibo Zhao, Yu Quan Chong, Chen Wang, Katia Sycara, Matthew Johnson-Roberson, Dhruv Batra, Xiaolong Wang, Sebastian Scherer, Zsolt Kira, Fei Xia, Yonatan Bisk. Paper | Website

Nov 2019 **Towards Zero-Shot Alignment and Retrieval for Forensic Detection**.
Vidhi Jain*, Amrit Setlur*, Aditya Pratapa*. Report

Tech Reports

Aug 2025 **Lowering Barriers in Human-Robot Communication.** PhD Thesis. Report

Aug 2021 **Towards Explainable Embodied AI.** Masters Thesis. Report

Jun 2018 **Investigating the viability of Generative Models for Novelty Detection.** Bachelors Thesis. Report

Selected Awards and Honors

Dec 2025 **BITS Global 30 Under 30 Award** in Research leaders – Academia category Link

Aug 2020 **CIFAR Deep Learning Reinforcement Learning Summer School**

Apr 2020 **Heidelberg Laureate Forum (HLF '20)** selected as a Young Researcher among 224 others worldwide.

Jul 2019 **J N Tata Endowment Scholarship for Higher Studies, India**

Jun 2019 **K. C. Mahindra Scholarships for Post-Graduate Studies Abroad**

May 2017 **ACM-W Scholarship for Attendance at Research Conferences**

Mar 2017 **Citi Women Leader Award (CWLA), Mumbai, India**, Awarded one year of study scholarship covering upto INR 400,000 i.e. USD 6000, among the top 3 out of 1200 applicants

Jun 2016 **GE Foundation Scholar-Leaders Program (GEFSLP)**, Awarded USD 2000 for two years of study, among top 5 scholars selected in India

Mar 2016 **National Finalists at Microsoft's 'Build the Shield', Hyderabad, India**, Ranked 15th nationwide

2014–15 **Merit Scholarship** by Dean, BITS Pilani, *Top 1% among the batch of about 900 students*

2013 **Awarded Kishore Vaigyanik Protsahan Yojna (KVPY) Fellowship** by Dept of Science and Tech., Govt.of India

2010 **Awarded National Talent Search Exam (NTSE) Scholarship** by NCERT, India

2010 Ranked Top 1% in Regional State in **National Standard Examination in Junior Science (NSEJS)**

Talks

Mar 2024 ***How to think of multimodality in Embodied AI and Robotic Manipulation?***.
Guest Lecture in Multimodal Machine Learning course. Lecture Video

Apr 2023 ***Training vision models in PyTorch***.
Hands-on lecture in Computer Vision course. Lecture Video

Oct 2021 ***Three brushes to paint your research canvas***.
SAiDL (Virtual) Video | Event Poster

May 2019 ***Tutorial on Deep Learning with PyTorch***.
PyLadies Bangalore, India. Presentation | Event Poster

Jan 2019 ***The One in Asankhya Project***.
by Sukriti Paul and Mansi Goyal (Virtual) Video | Blog

Jul 2017 ***Invited panelist for 'Innovation by Young India'***.
telecasted on national news channel NDTV India at New Delhi office. Video

Professional Service and Leadership

Sep'23–Present **Reviewer** at IEEE International Conference on Robotics and Automation (ICRA), International Conference on Intelligent Robots and Systems (IROS)

Jan'20–Present **Reviewer** at International Conference on Machine Learning (ICML), Neural Information Processing and Signal (NeurIPS) and International Conference on Learning Representations (ICLR).

Jun'20–Jul'20 **Workshop Session Organizer** | Coping with Sample Inefficiency in Deep Reinforcement Learning (DRL) in embodied AI, ICML'20 WiML Un-Workshop, Virtual. Report | Presentation

Sep'17–May'18 **Co-Founder and Vice Chair** | BITS Pilani Association for Computing Machinery-Women (ACM-W) Chapter, *Started with 16 on-campus girls, organized project brainstorming and mentoring sessions*

May'15–Aug'15 **Technical Volunteer** | ARISE Impact, non-profit organization *Developed Android application for Audio-based learning for visually challenged*

Independent Research Supervision and Mentorship

Feb 2024 **Soumya Teotia** *(Senior thesis + intern)* Multi-task instruction following in Habitat simulator and developing code for easy transfer to physical robot. → **Masters at CMU RI**

May 2024 **Su Li** *(CMU sophomore)* Social teleoperation with VR headset and Hello Robot Stretch. Presented live demo at CoRL 2024, Munich, Germany.

Aug 2024 **Ashish Marisetty** *(CMU MIIS graduate for independent study)* Language-conditioned video generation for high-level robot dynamics modeling. → **Intern at Nvidia**

Teaching

Jan'24-May'24 **Teaching Assistant** | 11777 Multimodal Machine Learning by Yonatan Bisk. Mentoring 5 teams with 3-5 students each for semester-long research projects.

Jan'23–May'23 **Teaching Assistant** | 16720 Computer Vision by Deva Ramanan. for 106 students enrolled in the course, prepared homework, and graded.

Jan'17–May'17 **Teaching Assistant** | CS F111 Computer Programming by Vishal Gupta, *Mentored 35 students for 4 hours per week for Code programming lab sessions.*

Aug'14–May'15 **Teaching Volunteer** | Computer Literacy Programme, National Service Scheme, *Tutored 12-15 rural natives for 3 hours per week on computer applications.*

References

Yonatan Bisk, Assistant Professor, Carnegie Mellon University, ybisk at andrew.cmu.edu

Debidatta Dwibedi, Research Scientist, Google DeepMind, Mountain View, debidatta at google.com

Akshara Rai, Research Scientist, Meta (previously Facebook), Menlo Park, akshararai at meta.com